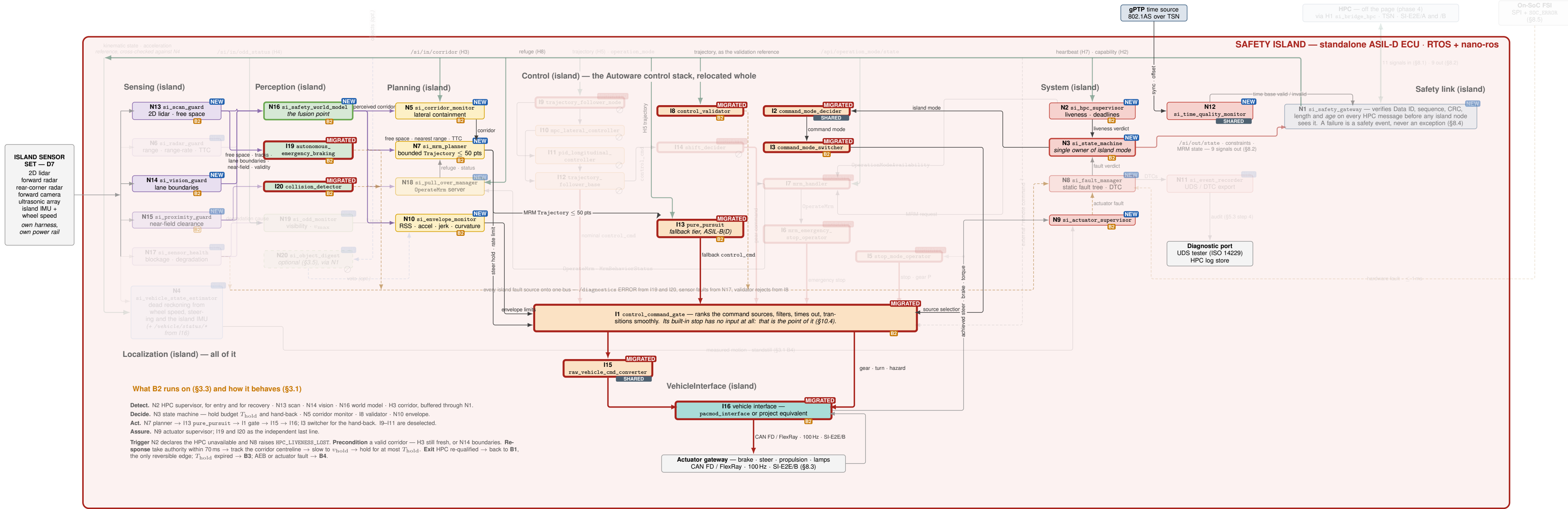
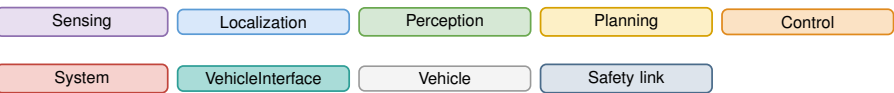


Phase 5 · B2 — Keep the vehicle in its lane after loss of the HPC

When the HPC is unavailable and B1 therefore cannot be used, take authority and hold the vehicle in its current lane — for a bounded window — while the lane is still observable.
The phase-5 drawing, unchanged: what B2 runs on — and the SHARED nodes every behaviour runs through — is in full colour, what it merely passes through is half-tone, and the rest of the island is faded.



Stacks — the colours of ArchitectureMain.pdf



Where each node came from

MIGRATED
the same Autoware node, migrated

NEW
new — does not exist in Autoware

18 migrated + 19 new = 37, plus the optional N20. Not drawn: I4, a header-only library with no topics, and I17 + I18, the legacy alternative of §4.3. I10 and I11 are plugins inside I9 and publish no topic of their own. ↑ the external / remote command reaches the gate (§5.2, §10.4) but has no row in the §8.1 table.

Behaviours each node enables (§3.3)

one fixed slot per behaviour
B2

SHARED

B2

Used by no behaviour

B1 degrade the ADS · B2 lane-keep hold after HPC loss · B3 pull over · B4 blind emergency stop. A filled tag means the node appears in that behaviour's Detect, Decide, Act or Assure cell of §3.3.

SHARED — every behaviour runs through it, so §3.3 names it against none: I2 turns island mode into command mode, I15 converts whichever command wins the gate, N12 qualifies the time base behind every age and deadline check.

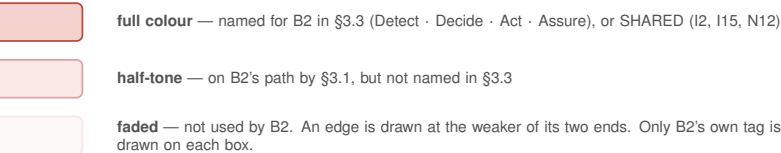
Hollow tag — not in §3.3, but §3.1 puts the node on that behaviour's path: N15 feeds I20, B2's independent last line, and is the near-field check behind B3's success criterion.

Used by no behaviour — I9–I12 and I14 track the HPC's nominal trajectory, which B2–B4 deselect and B1 only constrains. N20 is optional (§3.5) and required by none.

Messages



In this view — B2, lane-keep hold



Every edge on this page, with the design-document section it is taken from, is catalogued in ../messageflow/, which draws the same 37 nodes coloured by execution partition instead.